

Bootstrap Stack — BOOT Rack & Platform Bring-Up

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Companion to [SPEC.md §2A](#) (BOOT rack), [§7](#) (Kubernetes), [§6.7](#) (archive bus), [§11](#) (observability).

Code: [bootstrap/](#) · **Interactive walkthrough:** [demo/](#) (browser simulator of this exact process)

1 Scope

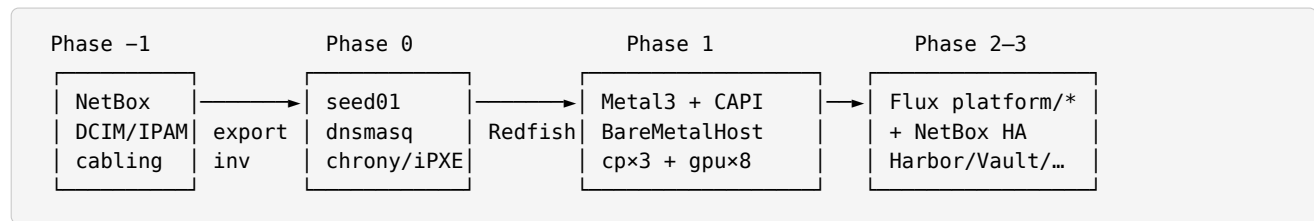
In scope (BOOT day-1)	Out of scope (later docs)
NetBox DCIM/IPAM/cabling SoT	Rail RoCE QoS detail → docs/network.md
Seed host DHCP/DNS/NTP/iPXE (from NetBox)	Hot FS CSI product choice → docs/storage.md
Metal3 + CAPI bare-metal	Full BOM SKUs → docs/bom.md
3-node K8s control plane	Facility elevations → docs/facility.md
Flux GitOps root	
Harbor, Vault, ESO, NetBox HA	
Prometheus / Grafana / Loki / Alloy	
Redpanda inference bus	
NVIDIA GPU Operator + Network Operator	
KServe (RawDeployment) + KubeRay operator	

2 Physical placement (BOOT)

RU role	Qty	Notes
seed01 (bootstrap utility)	1	First boot; may later become util or stay out-of-band
Control plane cp01–cp03	3	32c / 256 GB / NVMe (SPEC §7.1)
Utility util01–03	3	Harbor etcd/DB, Vault raft, Redpanda, Loki backends if not on STOR
Spines 7060DX5-64S	2	Prefer BOOT (SPEC §2A)
Console / jump	1	

GPU workers **never** share BOOT power domain.

3 Logical architecture



See [docs/netbox.md](#) and [docs/cabling.md](#). **Management cluster:** `clusterctl` runs against a small k3s (or existing) on seed01. Workload cluster `ai-cluster` is the production plane; move CAPI management to a dedicated util node after day-1 if desired.

4 Addressing (defaults)

IPAM lives in NetBox. Generated snapshot: `bootstrap/inventory/ipam.yaml`.

Device inventory snapshot: `bootstrap/inventory/cluster.yaml` (**GENERATED** — `bash bootstrap/scripts/netbox-sync.sh`).

Plane	CIDR	Example
Mgmt	10.10.0.0/24	seed 10.10.0.10, API VIP 10.10.0.20
OOB/BMC	10.20.0.0/24	via 7010TX-48
Pod	10.244.0.0/16	
Service	10.96.0.0/12	

4.1 Domain default: **ai.local** (replace with enterprise DNS).

5 Procedure

5.1 Prerequisites

- Ubuntu 24.04 on seed01, dual-homed to mgmt + reachability to OOB
- Operator workstation: `kubectl`, `clusterctl`, `flux`, `yq`, `jq`
- BMC Redfish on all servers; credentials via env (`BMC_USERNAME`, `BMC_PASSWORD`)
- Ubuntu 24.04 cloud image + Ironic IPA assets under `http://seed/images/`
- Git remote for this monorepo (update `GitRepository` URL)

5.2 Commands

```
# seed01 – NetBox SoT first
cd bootstrap/netbox && docker compose up -d
export NETBOX_URL=http://127.0.0.1:8081
export NETBOX_TOKEN=$(bash scripts/create_token.sh)
python3 scripts/import_seed.py
bash ../scripts/netbox-sync.sh

cd ..
sudo bash scripts/00-seed-host.sh

# install k3s on seed for CAPI mgmt (example)
curl -sfL https://get.k3s.io | sh -
export KUBECONFIG=/etc/rancher/k3s/k3s.yaml

bash scripts/01-install-capi-metal3.sh
export BMC_USERNAME=... BMC_PASSWORD=...
bash scripts/02-apply-cluster.sh

# wait Ready
clusterctl describe cluster ai-cluster -n metal3
clusterctl get kubeconfig ai-cluster -n metal3 > ../kubeconfig-ai-cluster
export KUBECONFIG=$PWD/../../kubeconfig-ai-cluster

# CNI: install Cilium or Calico before Nodes Ready if not in Machine bootstrap
# e.g. cilium install

GIT_URL=https://git.example.com/org/super.git bash scripts/03-install-flux.sh
bash scripts/04-smoke.sh
```

5.3 Flux reconcile order

1. platform-core — cert-manager, node policy
 2. platform-security — Vault HA + External Secrets
 3. platform-netbox — production NetBox (DCIM/IPAM SoT)
 4. platform-registry — Harbor
 5. platform-observability — kube-prometheus-stack, Loki, Alloy
 6. platform-bus — Redpanda + inference topics
 7. platform-gpu — GPU Operator, Network Operator, RuntimeClass
 8. platform-serving — KServe, KubeRay, `models-rwx` PVC stub
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6 GPU path notes

- Worker Machines taint `nvidia.com/gpu=true:NoSchedule` until GPU Operator labels capacity.
 - Network Operator deploys OFED + RDMA device plugin; **per-rail** `SriovNetworkNodePolicy` is site-specific (8× ConnectX-7/BF3) — finalize in `docs/network.md`.
 - NCCL defaults ConfigMap in `platform/apps/gpu` — set `NCCL_IB_HCA` after device names stabilize.
 - Do **not** bake multi-TB weights into images; mount `models-rwx` (SPEC §6.6).
-

7 Security baseline

Control	Implementation
Secrets	Vault raft HA; apps via External Secrets
Registry	Harbor + TLS; mirror NGC/vLLM images
Bootstrap secrets	BMC creds only in env → generated Secret (not git)
API	Control plane audit logs on; cert-manager issuers
Tenancy later	Gatekeeper/Kyverno + namespaces per team

Rotate `harborAdminPassword` / Grafana admin on first login.

8 Inference archive integration

Redpanda topics (`inference.records`, DLQ, tombstones) buffer async capture before Parquet → object lake (SPEC §6.7). Compactor Deployments join when STOR S3 endpoint exists.

9 Acceptance criteria

- ☐ seed DHCP/DNS answers for all inventory hosts
 - ☐ NetBox export = live inventory; cabling 64+64+BMC
 - ☐ 3 CP nodes Ready; API VIP serves **6443**
 - ☐ 8 GPU workers Ready; [nvidia.com/gpu](https://www.nvidia.com/en-us/gpu) Capacity = 8 each (64 total)
 - ☐ DCGM exporter scraped by Prometheus
 - ☐ Harbor push/pull from GPU node
 - ☐ Vault unsealed / raft peers = 3
 - ☐ Redpanda Kafka API reachable from inference namespace
 - ☐ Example InferenceService applies (weights optional for dry-run)
-

10 Failure / recovery

Event	Action
seed01 loss	Static leases + images should be mirrored to util01; rebuild from scripts
Single CP loss	etcd quorum remains; replace BareMetalHost
Flux drift	<code>flux reconcile ks platform-gpu --with-source</code>
GPU driver break	GPU Operator operands reconcile; drain one rack leaf pair at a time

11 Next documents

0. [docs/build-guide.md](#) — **physical build first** (rack, power, cable, switch/server bring-up, burn-in)
 1. [docs/network.md](#) — EOS RoCE, EVPN, rail SR-IOV (consume NetBox IPAM)
 2. [docs/storage.md](#) — CSI for `models-rwx` + RGW lake
 3. [docs/k8s.md](#) — day-2 ops, upgrades, multi-tenancy
 4. [docs/bom.md](#) — exact BOOT server SKUs
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12 Revision

Version	Date	Notes
0.1	2026-07-18	Initial bootstrap stack